



Published in final edited form as:

Int J Pharm. 2023 May 10; 638: 122924. doi:10.1016/j.ijpharm.2023.122924.

Nucleic acid immunotherapeutics and vaccines: a promising approach to glioblastoma multiforme treatment

Lauren Dain¹, Guizhi Zhu^{1,2,*}

¹Department of Pharmaceutics and Center for Pharmaceutical Engineering and Sciences, Institute for Structural Biology and Drug Discovery, School of Pharmacy; The Developmental Therapeutics Program, Massey Cancer Center; Virginia Commonwealth University, Richmond, VA 23298, USA

²Department of Pharmaceutical Sciences, College of Pharmacy, University of Michigan, Ann Arbor, MI 48109, USA

Abstract

Glioblastoma multiforme (GBM) is a deadly and difficult to treat primary brain tumor for which satisfactory therapeutics have yet to be discovered. While cancer immunotherapeutics, such as immune checkpoint inhibitors, have successfully improved the treatment of some other types of cancer, the poorly immunogenic GBM tumor cells and the immunosuppressive GBM tumor microenvironment have made it difficult to develop GBM immunotherapeutics. Nucleic acids therapeutics and vaccines, particularly those of mRNA, have become a popular field of research in recent years. This review presents the progress of nucleic acid therapeutics and vaccines for GBM and briefly covers some representative delivery methods of nucleic acids to the central nervous system (CNS) for GBM therapy.

Graphic Abstract

*Corresponding author: G.Z. (email: guizhiz@umich.edu).

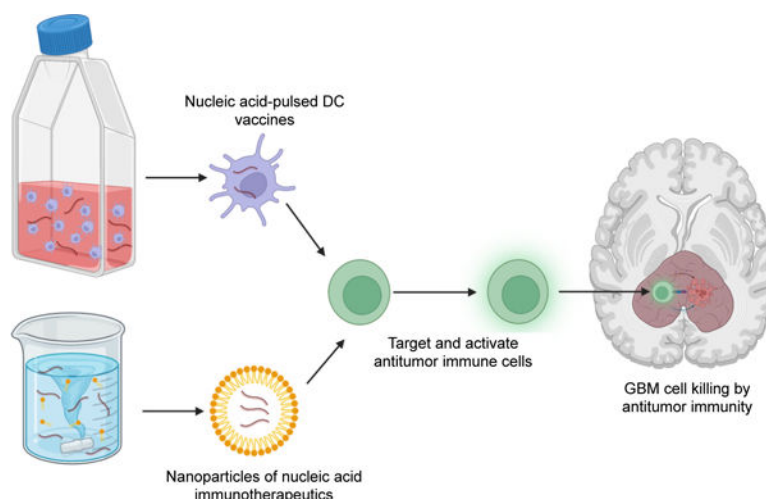
Publisher's Disclaimer: This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

Conflicts of Interest

The authors have no conflicts of interest.

Declaration of interests

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Guizhi Zhu reports financial support was provided by National Institutes of Health. Guizhi Zhu reports financial support was provided by American Cancer Society. Guizhi Zhu reports financial support was provided by US Department of Defense.



Keywords

nucleic acid vaccine; nucleic acid immunotherapeutics; glioblastoma immunotherapy; lipid nanoparticles; mRNA vaccine

1. Introduction and Background

GBM is the most aggressive and lethal CNS tumor [1, 2], affecting both adult and pediatric populations [1, 3]. The current standard of care for GBM is surgery, radiotherapy, and chemotherapy, specifically temozolomide. Unfortunately, with these current treatments, GBM patients still do not receive much survival benefit with the median overall survival (OS) at approximately 15 months [4, 5]. In fact, the prognosis for GBM patients has not improved much in the past four decades [2], likely because of several complications. For example, GBM consists of malignant astrocytes, which are highly infiltrative and essentially guarantee incomplete resection of the tumor [4]. Moreover, GBM is often highly immunosuppressive, which hampers the development of anti-GBM immunity and inhibits the infiltration of anti-GBM immune cells. In general, GBM tumors also have very low mutation burdens. Yet, high mutation burdens resulting in high numbers of the corresponding tumor neoantigen-specific T cells are important for successful responses to immunotherapy such as immune checkpoint inhibitors (ICIs) [6], which may explain the lack of progress in clinical trials testing ICI therapy in GBM. Finally, the blood-brain barrier (BBB) and blood-tumor barrier (BTB) in GBM represent physiological barriers against effective delivery of therapeutics to GBM tissues from the systemic circulation. This leaves the field of GBM treatment desperate for new advances that may finally improve the prognosis for GBM patients.

As mentioned above, the current standard standard of care for GBM is surgery, radiotherapy, and temozolomide. The first mentions of glioma tumors were found at the turn of the 19th century in British autopsy reports [7]. In 1865, the German pathologist Rudolf Virchow provided the first comprehensive histomorphological description, showed that tumor cells were malignant transformations of normal CNS glial cells, and first coined

the term “glioma” [7]. Surgery was the only option for removing tumors until The Brain Tumor Study Group (later renamed The Brain Tumor Cooperative Group) established post-operative radiation therapy as the new standard of care for the treatment of GBM through the success of three randomized clinical trials in the early 1970s [8]. In 2005, the European Organization for Research and Treatment of Cancer (EORTC) hosted a trial showing improved overall survival of GBM patients who received temozolomide as well as radiation therapy as a post-operative adjuvant when compared to those who received radiation alone [9], an effect that was still appreciated years later [10]. Improvement to surgical technique came in 2006 when it was shown that more maximal cytoreductive surgery could be achieved with a fluorescence-guided technique utilizing 5-aminolevulinic acid [11], which provided an improved six-month progression free survival compared to control patients. Even so, there is still much improvement to be made as median OS still stays at approximately 15 months.

The brain has been historically considered an immunoprivileged organ as the formidable BBB prevents the passage of molecules and materials from the systemic bloodstream into the brain. A more recent study, though, has illustrated that the brain is actually not immunoprivileged since it can communicate with the systemic immune system via lymphatic channels within the meninges [12], and a variety of immune cells and molecules are able to access the CNS region. This opened the door to attempt to utilize the systemic immune system to target and kill brain tumor cells.

Many factors play a role in the difficulty of treating GBM with immunotherapy. One of the most complex obstacles is the GBM tumor microenvironment, reviewed by Charles et al. [13]. Another is that GBM tumors are immunologically “cold,” meaning that the tumor microenvironment is so immunosuppressive that it inhibits the elicitation of antitumor immunity and prevents the effective infiltration of antitumor cells and molecules. GBM tumors are immunosuppressive [5, 14] by releasing immunosuppressive cytokines and growth factors such as interleukin-10 (IL-10), prostaglandin E2 (PGE-2), transforming growth factor beta (TGF- β) [15], and vascular endothelial growth factor (VEGF) [14]. Foundational studies in glioma immunology have also illustrated the dysfunction of cell-mediated immunity in GBM patients [16–18]. Glioma cells are also poor at presenting tumor antigens because GBM cells notably downregulated the expression of B7 costimulatory molecules, which are required for T cell activation [19]. In addition, tumor associated macrophages (TAMs) play a critical role in the immunosuppressive nature of the GBM microenvironment. The immunosuppressive M2-like macrophages, which make up the majority of non-neoplastic cells in GBM tumors [20], decrease antitumor inflammation and exhibit protumoral effects such as the promotion of tumor growth and invasion. Taken together, these challenges make the development of a successful immunotherapeutic for GBM no small feat.

Nucleic acid therapeutics have already become a popular topic for both preclinical and clinical cancer therapy research. Studies exploring nucleic acids as therapeutics (mostly for gene knockdown and protein replacement) in various pediatric cancers such as hepatoblastoma [21–23], nephroblastoma [24, 25], neuroblastoma [26, 27], osteosarcoma [28, 29], medulloblastoma [30–32], retinoblastoma [33, 34], and acute lymphoblastic

leukemia (ALL) [35, 36] have been largely successful and promising. Several clinical trials since 1999 investigating the use of antisense oligonucleotides (ASO) have been undertaken (listed in [37]), though few, if any, have listed results. This lag in progress is likely due to the difficulties with ASO delivery to the brain [3] and affordable cost of production. For GBM, mRNA has also recently been explored as a new method for gene therapy delivery [38–40].

Along with gene therapy, gene knockdown, and protein replacement, nucleic acids have also begun to be tested as immunotherapeutics for cancer. Several clinical trials using RNA vaccines for various types of cancer have been completed and/or are still underway, illustrating the excitement for this relatively new field of study. These clinical trials can be found searching ‘RNA Vaccine’ and ‘Cancer’ at www.clinicaltrials.gov. Besides vaccines, aptamers, immunostimulatory nucleic acids, gene-regulating nucleic acids, gene-editing nucleic acids, and plasmids have also shown promise as new classes for cancer immunotherapy (previously reviewed by our lab in [41]).

Due to the success of nucleic acid therapeutics for other types of cancer, there has arisen a hope that this new field of study might also be beneficial for the treatment of GBM. One group that has been greatly researched is the use of oligonucleotides (OTs) (extensively reviewed in [3]). One example of the usefulness of OTs is as enzyme inhibitors. It has been previously shown that the reactivation of telomerase, an enzyme that prevents the shortening of telomeres during genome replication, is a hallmark of cancer survival and progression, making it a popular target for cancer therapeutics [42]. Marian et al. showed that using the OT telomerase inhibitor imetelstat successfully decreased tumor cell proliferation and tumor growth both *in vitro* and *in vivo*; this drug was then later used in a phase II clinical trial with pediatric brain tumor patients [43]. Other forms of nucleic acids that have shown success as GBM treatments are small interfering RNA (siRNA) [44], micro RNA (miRNA) [45], aptamers [46], mRNA [47], DNA [48], and peptide nucleic acids (PNAs) [49], among others. Zhang et al. [50] reviewed the broader topic of the current state of nucleic acid therapeutics for brain tumors; the focus of this review is a brief overview of nucleic acid immunotherapeutics, specifically, for GBM (summarized in Table 1), the role of circular RNA (circRNA) in GBM and its potential as a GBM therapeutic, and an overview of nucleic acid delivery methods to the CNS and the systemic immune system as therapeutics, especially nucleic acids, rely on effective delivery to target tissues. Delivery systems used for the local and systemic delivery of nucleic acid immunotherapeutics for cancers, including GBM, are critical to protect the cargo from enzymatic degradation in the body, prolong the half-life and enhance the bioavailability of the nucleic acids, increase the delivery to target tissues and cells, and eventually increase the therapeutic efficacy and minimizing adverse side effects. Therefore, we also briefly discussed some representative delivery systems reported for nucleic acid immunotherapeutics of GBM.

2. The current classes of nucleic acid immunotherapy for GBM

2.1. mRNA-pulsed dendritic cell vaccines

Since the 1990s, tumor cell vaccines for cancer therapy have become quite popular. The possibility of not only besting the tumor’s immune suppression but also weaponizing the innate and adaptive immune responses toward anti-tumor immunity [51] could drastically

alter the way medicine fights even the toughest cancers such as GBM. Several types of vaccines have been created, such as peptide vaccines [52], mesenchymal stem cell vaccines [53], and viral antigen vaccines [54], but many utilize DCs, a type of antigen-presenting cell (APC). DCs are widely studied to develop tumor immunotherapeutics because they play a pivotal part in the activation of antitumor immunity.[55]

There are several reasons why mRNA is preferable as cancer therapeutic vaccines. Because patient-derived mRNA can be amplified *in vitro*, a relatively small number of cells is needed from which to derive mRNA [2, 55], which allows for the development of an mRNA-based vaccine for patients who only have a very small biopsy to work with [55]. mRNA also has an increased safety profile over DNA as there is no worry about integration into the host genome [2, 56]. The Ashley et al. group was among the first to show efficacy of an mRNA-pulsed DC vaccine in a mouse tumor model [55]. In experiments where mice were first vaccinated prophylactically then challenged with B16 melanoma cells second, and where mice had pre-existing B16 tumor and subsequently vaccinated, the animals that received the B16 total cell RNA vaccine had significantly prolonged survival over animals vaccinated with B16 tumor extract alone or SMA 560 (spontaneous astrocytoma cell line) extract or RNA [55]. As the first study illustrating the mRNA-pulsed DC vaccines for cancer therapy, this group opened the door to explore this technology for other cancer types. Melanoma and GBM, both deriving from neural ectoderm-descendent cells, share several tumor-associated antigens (TAAs) making this development promising for later exploration with GBM models.

More groups have been using mRNA encoding particular tumor antigens to elicit antigen-specific antitumor T cell responses rather than whole-cell mRNA. The receptor for hyaluronan-mediated motility (RHAMM) is an intracellular protein that has some function in locomotion [57] and has been linked to aberrant signaling through Ras and Ser and metastasis formation [58] and tumor progression in GBM [58] and other cancers, such as B cell malignancies [57]. Amano et al. [58] found that in mice with implanted KR158B tumors, mean mouse survival was significantly increased in mice vaccinated with RHAMM mRNA DC cells (mean survival 61.8 days) compared to mice vaccinated with nontransfected DCs (31.6 days) and mice without vaccination at all (25.6 days). The authors recognized that as there was only a twofold increase in mean survival days in animals treated with RHAMM mRNA DC vaccines, and hypothesized that RHAMM might not be immunogenic enough to induce complete tumor regression [58]. For a more robust and long-lasting anti-cancer immune response, CD4⁺ T helper cells must be activated alongside CD8⁺ cytotoxic T cells [59]. Indeed, Amano et al. found both CD8⁺ and CD4⁺ infiltration in the brain tumors of mice treated with RHAMM mRNA-transfected DCs whereas the mice who had received no treatment had no T cell infiltration at all [58]. While the survival benefit of the RHAMM mRNA DC vaccine was only modest, the finding of both CD8⁺ and CD4⁺ in the brain is a promising indication that mRNA DC vaccines are mounting the necessary immune response to have an anti-tumor effect when partnered with appropriate tumor antigens.

Similarly, Saka et al. [14] focused on creating a vaccine with mRNA encoding the protein IL13ra2, a subunit of the IL-13 receptor. This group found that in their GBM mouse

model using the KR158B cell line, the Il13ra2 protein was overexpressed specifically in glioma cells but absent in normal brain tissues. Their mice treated with the Il13ra2 mRNA-transfected DCs survived significantly longer than mice treated with nontransfected DCs and those without treatment. In fact, all mice that were treated with nontransfected DCs and those without treatment had died by experiment day 40 while 50% of the mice treated with Il13ra2 mRNA-transfected DCs were still alive at day 35 with a small portion of the cohort alive past experiment day 80. This group also found that their DC vaccine stimulated both CD8⁺ and CD4⁺ T cells as evidenced by their infiltration into and around tumor tissue [14]. While this preclinical data is promising, it has been shown that only a minority of GBM patients overexpress Il13ra2 in tumor cells [60] and because Il13ra2 is expressed in a myriad of other organs in humans [61, 62], multi-system toxicity may ensue with systemic delivery of Il13ra2 mRNA DC vaccine in humans [14].

A major issue with using a mouse model for immunotherapy research is discrepancies between human and murine immune systems. Do et al. [4] attempted to overcome this hurdle by creating a mouse model with a humanized immune system to test a DC vaccine pulsed with CD133 mRNA. CD133 has been previously shown to be a cell surface marker for brain tumor stem cells (BTSCs) [63] and is negatively correlated with patient survival [64]. The authors first showed that the CD133 protein was efficiently loaded onto both MHC class I and MHC class II for presentation to both CD8⁺ and CD4⁺ T cells and that the greatest number of CD133⁺ BTSCs were killed by CD8⁺ and CD4⁺ T cells stimulated with the modified human CD133 mRNA-transfected DC vaccine *in vitro*. They then illustrated successful humanization of their immune-deficient mice by detecting human CD8⁺ and CD4⁺ T cells four weeks after stem cell transplant. The authors then injected human GBM cells intracranially into the humanized mice. The mice subsequently treated with the modified human CD133 mRNA-transfected DC vaccine had a significantly increased survival with all mice in this group alive greater than 60 days, while mice treated with nontransfected DCs had a median survival of 40 days and control mice treated with PBS had a median survival of 38 days [4]. This work highlights the utility of the CD133 mRNA-transfected DC vaccine as well as the utility of a (partially) humanized mouse model to closely resemble human patients in preclinical studies.

Vik-Mo et al. [2] brought a glioma stem cell (GSC) DC vaccine to clinical trial using autologous GSC mRNA (protocol detailed in Figure 1). Because GBM tumors are incredibly heterogeneous, within each tumor and between patients, using autologous whole-cell GSC mRNA will hopefully stimulate each patient's T cells toward their unique array of antigens rather than trying to create a one-size-fits-all therapy with specific antigens that may not be expressed by all tumor cells or in all patients. In this study, the tumors of GBM patients were biopsied and stem cells were allowed to grow in culture. Once tertiary tumorspheres had grown, cells were harvested, whole-cell GSC mRNA was collected, and DCs were transfected. Once patients had received standard surgical and radiotherapy, one group received the DC vaccine alongside standard temozolomide treatment while the control group received standard therapy only. The authors found that the vaccine-treated group had a significantly longer progression-free survival (median 694 days) compared to the matched control group (median 236 days). All other endpoints, including overall survival,

were not significant [2]. However, the results produced by this group provide some hope that mRNA-transfected DC vaccines may continue to show promise in clinical trials.

mRNA-transfected DC vaccines have also shown favorable safety in both preclinical and clinical studies. Caruso et al. [1] found that none of their DC vaccine-treated patients developed autoantibodies or any other evidence of autoimmune disease. Amano et al. [58] and Saka et al. [14] were both unable to detect any toxicity in the brain tissue of their DC-treated animals using histological studies. And Vik-Mo et al. [2] found no clinical evidence of cerebral edema or autoimmune encephalitis, retinal or uveal inflammation, or autoimmune reactions against stem cells, all potentially organ- or life-threatening complications.

2.2. Additional RNA therapies

In addition to DC vaccines, a variety of vehicles have been explored for vaccine delivery. Sayour et al. [65] proposed that DC vaccines may not be the best choice as DC expansion and collection post-leukaphoresis can take several weeks, and this can cause “a significant delay for patients with aggressive malignancies and may preclude candidates from trial with rapid disease progression” [65]. Instead, they identified a nanoliposomal formulation, 1,2-dioleoyl-3-trimethylammonium-propane (DOTAP), that can load tumor antigen mRNA and be administered as a vaccine. The RNA-nanoliposome complex was referred to as the RNA-NP. The authors showed that the whole-cell RNA-NPs did have an antitumor effect on survival and tumor volume in mice with orthotopic GBM tumors, but the effect on survival was not significantly different from that of the DC formulation of the same vaccine, which were both significantly increased when compared to untreated mice [65]. A later study from the same group showed that this NP conjugated to a small interfering RNA (siRNA) targeting PD-L1 significantly decreased PD-L1 expression in GBM-associated myeloid cells and claimed increased survivorship when combined with PD-1 checkpoint blockade in a largely fatal GBM model using GL261 cells [66].

Other groups have used RNA NPs not to target tumor cells directly, but to modulate the immune system. It has been previously shown that signal transducer and activator of transcription 3 (STAT3) contributes to malignancy by inhibiting the expression of several proinflammatory cytokines and upregulating VEGF and IL-10, and that downregulation of STAT3 can initiate immune-mediated antitumor effects in both glioma animal models and patients [67, 68]. Yaghi et al. [69] sought to target STAT3 with their newly developed miRNA-NP technology. This group previously found that miR-124 plays an important role in STAT3 regulation, and that administering miR-124 has antitumor effects in preclinical late stage glioma models using the lipid-mediated transfection reagent Lipofectamine [70]. The present study [69] compared the therapeutic and targeting efficacies between the previously used Lipofectamine formulation of miR-124 (miR-124 + Lipofectamine) and a new miR-124 NP (named LUNAR-301) formulation using Arcturus Therapeutics’ lipid-enabled and unlocked nucleic acid modified RNA (LUNAR) technology, for which the proposed mechanism of action is shown in Figure 2. Lipofectamine is a common transfection reagent that aids in the delivery of RNA and DNA, primarily in cultured cells, by creating an overall positively charged liposome that can then be taken up by cells. The LUNAR

technology explored by Yaghi et al. was an attempt to improve upon this delivery method through the use of a proprietary LNP rather than Lipofectamine. The authors first tested the therapeutic efficacy of four experimental miR-124 NPs and miR-124 + Lipofectamine. Mice with intracerebral GL261 tumors treated with NB5-55-2 (subsequently referred to as LUNAR-301) demonstrated the greatest survival time with >60% of animals surviving past 70 days. The three remaining NP formulations and miR-124 + Lipofectamine had median survivals of 30.5, 43.5, 6, and 40 days, respectively. Targeting efficacy was elucidated by quantifying miR-124 concentration in peripheral blood mononuclear cells (PBMCs). miR-124 delivered via LUNAR-301 reached significantly higher concentrations in PBMCs than did miR-124 delivered via Lipofectamine, and maintained this significantly higher concentration for >24 hours. LUNAR-301 was also found to target CD11b⁺ monocytes and macrophages, which are significant cell subsets for STAT3 regulation. The promising results of this study and the commercial availability of the LUNAR technology make this an encouraging avenue of investigation.

As discussed previously, immune cells such as TAMs play a critical role in GBM survival and immunotherapeutic efficacy. Previous work has illustrated the opposing natures of M1 and M2 macrophages in a cancer setting with M1 cells being pro-inflammatory and anti-tumor, and M2 cells being immunosuppressive and pro-tumor [71, 72]. Zhang et al. [20] tested the hypothesis that RNA NPs could be utilized to reprogram M2-like TAMs to a more pro-inflammatory M1-like phenotype via gene delivery in mouse models of advanced-stage ovarian cancer, metastatic melanoma, and GBM. The RNAs that the authors included in their NPs were Interferon Regulatory Factor 5 (IRF5), which favors M1 polarization [73], and IKK β , a kinase that phosphorylates and activates IRF5 [74]. After confirming their technique for transfection (31.9% of macrophages were successfully transfected without compromising viability), the authors tested whether the delivery of IRF5 and IKK β RNA via NPs would increase levels of IRF5 translation; though gene expression wasn't lasting, the IRF5 protein remained upregulated through day 5 [20]. M2 macrophages were induced by culturing bone marrow derived macrophages (BMDMs) with IL-4 and then transfected with the IRF5/IKK β -NP. The authors found that genes classically expressed by M2 macrophages (such as Serpinb2 and Ccl11) were downregulated and important M1 genes (like Ccl5) were upregulated, evidencing the success of transforming M2 macrophages into a more M1-like phenotype. When used to treat a transgenic mouse model of PDGF β -driven glioma (PDG mice [75]), the IRF5/IKK β -NPs only provided a 5-day increase in survival over untreated controls, but combining IRF5/IKK β -NP therapy with radiotherapy resulted in an average survival of 52 days compared to the PBS-treated control group, which had an average survival of 25 days. The authors concluded that IRF5/IKK β -NP treatment in conjunction with standard of care for GBM may confer some survival benefit for GBM patients for which other therapies have failed [20].

2.3. DNA therapies

As DNA vaccines have been explored as a treatment option for various infectious diseases [76] and other cancers [77], it is reasonable that researchers have also been trying to develop DNA vaccines for GBM. DNA vaccines are relatively inexpensive to produce and are more

stable than linear mRNA, which makes them an attractive option [78], though they have the potential to integrate into the host genome [2, 56], which detracts from their appeal.

As mentioned previously, melanoma and GBM have many TAAs in common as they are both derived from the neural ectoderm. Several TAAs for melanoma (deemed melanoma-associated antigens or MAAs) have been discovered including the melanoma antigen gene family (MAGE), the B-M Antigen-1 family (BAGE), and the G Antigen family (GAGE), gp100, tyrosinase, and tyrosinase-related proteins (TRP) 1 and 2 [79, 80]. Two of the antigens shared between melanoma and GBM is TRP-2_{180–188} and gp100_{44–59} [with insertion positions at (18) and (191) and summarized in Table 2], which are a CD8 epitope and a CD4 epitope, respectively [78]. With this information, Lopes et al. [78] sought to create a DNA vaccine that harnessed the immunogenicity of vesicular stomatitis virus glycoprotein (VSV-G) with the insertion of the TRP-2_{180–188} and gp100_{44–59} T cell epitopes, a design they named plasmid to deliver T cell epitopes (pTOP) seen in Figure 3. The authors first evaluated whether pTOP was efficacious in killing subcutaneously implanted GL261 cells. They found that the tumor volumes of mice vaccinated with pTOP on days 2, 9, and 16 after tumor cell implantation had significantly smaller tumors than the untreated controls, and 6/7 vaccinated mice lived >80 days and considered long-term survivors, compared to 1/7 mice living >80 days in the untreated group [78]. The authors next wanted to test whether pTOP could treat mice with intracranially implanted tumors. They found that while pTOP treatment alone or tumor resection alone could not protect mice from disease progression, significant improvement in survival was achieved when pTOP was combined with bulk tumor resection with 78% of mice surviving >250 days [78]. From these results, the authors concluded that combining a DNA vaccine such as pTOP with surgical resection, which is already part of GBM standard of care, provides a new avenue of cancer treatment that should be explored further.

3. circRNA for GBM

circRNAs were first discovered in the early 1990s [81]. They were originally thought to be noncoding, but several studies have shown that some circRNAs do indeed encode and translate proteins. As they lack 5' caps and poly-A 3' tails and therefore cannot be cleaved by exonucleases or poly(A)-specific ribonuclease (PARN) [82, 83], circRNAs are much more stable than their equivalent linear mRNA [84, 85]. More recently, studies have found that some circRNAs play important roles in tumorigenesis and other pro-tumor activities in several types of cancer (reviewed in [86]), including GBM. Wang et al. [87] found that circRNA matrix metalloproteinase-9 (MMP-9), or circMMP9, formed from exons 12 and 13 of MMP9 (Figure 4), may have oncogenic properties in GBM. Overexpression of circMMP9 in U87 cells, which poorly express circMMP9 endogenously, was associated with a significant increase in proliferation, migration, and invasion of the U87 cells *in vitro*. On the other hand, silencing of circMMP9 in U251 cells, which highly express circMMP9 endogenously, was associated with a significant decrease in the migration and invasion of U251 cells *in vitro*. In mice, overexpression of circMMP9 was associated with a marked increase in the growth rate of U87 subcutaneous xenografts while silencing of circMMP9 was associated with a marked decrease in the growth rate of U251 subcutaneous xenografts, lending the authors to believe that this circRNA may be able to be exploited for GBM

therapy in the future [87]. As endogenous circRNAs and their functions have moved into the limelight, many other groups have discovered how these circRNAs directly or indirectly (through the proteins they encode) affect GBM tumorigenicity and patient prognosis [88–97]. Targeting the pro-tumor circRNAs and/or harnessing the anti-tumor circRNAs could prove to be a successful new field of GBM therapeutics (as reviewed in [98]).

With the ability to synthesize custom RNA sequences, it follows that this technology could be taken advantage of to create unique treatment options for GBM patients, as well as patients with other diseases. Several studies have shown that synthetic circRNAs have promising therapeutic potential as onco-miRNA sponges in gastric cell carcinoma [84], esophageal carcinoma [85], and lung adenocarcinoma [99], among others. But circRNAs are not only destined to be miRNA sponges. CircRNAs can translate proteins if an IRES (internal ribosome entry site) is placed in front of an open reading frame [100]. As such, circRNAs can also function as vaccines, for cancer or otherwise, if the circRNA encodes an antigenic protein. For example, a circRNA vaccine against SARS-CoV-2 was reported [101], and prototypical circRNA vaccines for cancer were explored [102]. The number of groups exploring the oncotherapeutic potential of linear mRNA and circular RNA vaccines is rapidly expanding, and the field is likely to see incredible progress over the next several years.

The utilization of circRNAs to treat disease is still a dawning field. Thus far, one of the biggest challenges for circRNA therapies is the engineering of efficient protein/peptide-expressing circRNA and the reproducible large-scale manufacturing of circRNA that will be needed to move preclinical successes into clinical trials. There are several methods for circularizing linear mRNA with varying degrees of success [103]. Another challenge is that unlike controlled preclinical studies using relatively homogeneous disease animal models, human tumors are not so homogenous. As a step toward address this challenge, personalized medicine, such as patient-specific personalized circRNA vaccines, can be developed based on the bona fide tumor antigens identified from each patient. Overall, circRNA cancer immunotherapeutics and vaccines may provide a breakthrough for successfully treating GBM patients.

4. Brief overview of nucleic acid delivery methods

Optimal prophylactic or therapeutic efficacies of nucleic acid vaccines and therapeutics require their delivery to target tissues and cells. Nucleic acids in particular do not fair well in the body without the protection of a delivery system as they are degraded quickly by ubiquitous nucleases [104]. Nucleic acids alone can also trigger a local or systemic immune responses [105], which may lead to adverse side effects. Moreover, because nucleic acids are often relatively large and negatively charged, their entrance into cells is negligible without a carrier [104]. Though this topic has been extensively covered, this review will highlight a few representative delivery approaches (summarized in Figure 5).

The discussion about LNPs above focused on those studies that utilized LNPs for RNA vaccines for brain tumors, specifically. This section will focus more on LNPs as a delivery vessel targeting the brain or lymph nodes for various other diseases and disorders, mostly of

the CNS. Chen, et al [105] designed a LNP that targeted lymph nodes where their mRNA vaccine can best induce an immune response. They discovered that the specific structure and size of the LNP was crucial for its ability to target lymph nodes. They found that LNPs constructed of lipids with shorter tail lengths were most efficient at reaching and accumulating in the lymph nodes while lipids with longer tails exhibited almost no efficacy. Replacing the ester bond with an amide bond, or replacing the methyl groups of the amine head to a hydroxyl, ethyl, or *N*-(1,2-ethanediyl)acetamide group also significantly decreased the LNPs' lymph node-delivery ability. Further, they showed that a significant CD8⁺ T cell response was initiated when their LNPs were loaded with a model protein antigen chicken ovalbumin (OVA)-encoding mRNA vaccine [105]. This technology can be promising not only for the delivery of nucleic acid vaccines, but also the delivery of other small molecule immunotherapeutics.

As the BBB prevents the systemic penetration of a variety of molecules into the brain, scientists have needed to be creative with their strategies to deliver therapeutics to the brain. The same group as above [107] took advantage of the natural CNS tropism of neurotransmitters. Tryptamine derivatives, including dimethyltryptamine, have been previously shown to cross the BBB by active transport [110]. Ma et al. [107] used dimethyltryptamine to help deliver therapeutics into the brain by attaching dimethyltryptamine molecules to the exterior of their LNP. They also found that LNPs containing shorter tail lengths had greater delivery efficacy, akin to what the group described in the previous paragraph determined. With this LNP, they successfully delivered a small molecule drug (amphotericin B), an antisense oligonucleotide (Tau-ASO), and a gene editing protein (GFP-Cre) into the brains of mice via tail vein injection. Another molecule that is rapidly taken up by the brain is glucose, the brain's primary source of energy. Anraku et al. [108] utilized the phenomenon of incredible glucose uptake into the brain after a fast has been broken. Simply decorating the exterior of their NPs with glucose did not increase the NPs' delivery into the brain in free-feeding mice, but putting the mice on a one-day fast, injecting the LNP intravenously, and then injecting a 20%wt glucose solution via intraperitoneal (IP) administration 30 minutes after injection of the NPs significantly increased the NP delivery into the mouse brains [108]. Yet another group [109] took advantage of the brain-targeting nature of the rabies virus glycoprotein (RVG) to deliver their extracellular vesicles containing circSCMH1, a circRNA generated from the *Scm Polycomb Group Protein Homolog 1* (SCMH1) gene, which is decreased in patients with acute ischemic stroke, and mouse and nonhuman primate models of acute ischemic stroke. Combining their extracellular vesicles containing circSCMH1 with RVG resulted in successful delivery of the circRNA therapeutic into the brains of mice and rhesus monkeys modeling acute ischemic stroke.

Another logical but more invasive delivery option is intrathecal (IT) delivery, where the protective meninges covering the brain and spinal cord are punctured. Nabhan et al. [106] asked the question of whether LNPs containing frataxin (FXN) mRNA could be delivered to the dorsal root ganglia (DRG) in a mouse model of Friedreich's ataxia via IT delivery. Indeed, they found that when compared with IV delivery of the same LNPs, IT delivery at the level of the lower lumbar spine of mice had superior DRG localization [106]. Though IT delivery is generally avoided in the treatment of humans due to the risk

of CNS infection, lumbar punctures are still widely used for diagnostic purposes. This study highlights how accessing the CNS via the compartment containing the spinal cord might be a technique to be utilized for therapeutics in the future.

5. Discussion and future directions

The lack of progress for the development of suitable therapeutics for GBM is outrageously frustrating for patients and their families, along with the physicians and other healthcare providers working to rid patients of this lethal tumor. Though average survival was marginally increased with the introduction of temozolomide chemotherapy and post-surgical radiation therapy into the standard of care so far [2], it's obvious that we must pursue a new direction if we hope to improve the lives of this patient population.

The success of immune checkpoint blockade for the treatment of cancers like melanoma [111] put immunotherapy in the spotlight as an exciting new frontier to explore and hopefully discover successful therapies for other forms of cancer as well. Clinical trials for the use of immune checkpoint inhibitors for GBM have not shown the same promise as it has for some other cancers, likely due to the low mutation burdens, the fairly immunosuppressive nature of GBM tumor microenvironment, and the poorly immunogenic GBM cells as discussed above [5, 6, 13, 14], but nucleic acid cancer vaccines are emerging as a viable option. DNA vaccines have been studied [77, 78] but RNA, particularly mRNA, has been studied more robustly. mRNA-pulsed DCs have been commonly studied in the setting of GBM [1, 2, 4, 14, 55, 58] because of the efficiency with which DCs activate the immune system and the safety profile shown in both preclinical and clinical studies. As the functions, applications, and large-scale current good manufacturing practice (cGMP) manufacturing of circRNA are further explored, circRNA would be promising as immunotherapies for cancer, including GBM, in the coming years. The current and future work performed in this area will hopefully bring forth the breakthrough in GBM therapy that this patient population so greatly deserves.

Acknowledgements

G.Z. acknowledges funding support from National Institutes of Health (R01CA266981, R01AI168684, R35GM143014, R21NS114455), US Department of Defense CDMRP Breast Cancer Breakthrough Award Level II (BC210931/P1), and American Cancer Society Research Scholar Grant (RSG-22-055-01-IBCD). The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

References

1. Caruso DA, et al. , Results of a phase 1 study utilizing monocyte-derived dendritic cells pulsed with tumor RNA in children and young adults with brain cancer. *Neuro Oncol*, 2004. 6(3): p. 236–46. [PubMed: 15279716]
2. Vik-Mo EO, et al. , Therapeutic vaccination against autologous cancer stem cells with mRNA-transfected dendritic cells in patients with glioblastoma. *Cancer Immunol Immunother*, 2013. 62(9): p. 1499–509. [PubMed: 23817721]
3. Krichevsky AM and Uhlmann EJ, Oligonucleotide Therapeutics as a New Class of Drugs for Malignant Brain Tumors: Targeting mRNAs, Regulatory RNAs, Mutations, Combinations, and Beyond. *Neurotherapeutics*, 2019. 16(2): p. 319–347. [PubMed: 30644073]

4. Do ASS, et al. , CD133 mRNA-Loaded Dendritic Cell Vaccination Abrogates Glioma Stem Cell Propagation in Humanized Glioblastoma Mouse Model. *Mol Ther Oncolytics*, 2020. 18: p. 295–303. [PubMed: 32728617]
5. Liu Y, et al. , TGF-beta2 inhibition augments the effect of tumor vaccine and improves the survival of animals with pre-established brain tumors. *J Neurooncol*, 2007. 81(2): p. 149–62. [PubMed: 16941073]
6. Garg AD, et al. , Preclinical efficacy of immune-checkpoint monotherapy does not recapitulate corresponding biomarkers-based clinical predictions in glioblastoma. *Oncoimmunology*, 2017. 6(4): p. e1295903. [PubMed: 28507806]
7. Stoyanov GS and Dzhenkov DL, On the Concepts and History of Glioblastoma Multiforme - Morphology, Genetics and Epigenetics. *Folia Med (Plovdiv)*, 2018. 60(1): p. 48–66. [PubMed: 29668458]
8. Barani IJ and Larson DA, Radiation therapy of glioblastoma. *Cancer Treat Res*, 2015. 163: p. 49–73. [PubMed: 25468225]
9. Stupp R, et al. , Radiotherapy plus Concomitant and Adjuvant Temozolomide for Glioblastoma. *The New England Journal of Medicine*, 2005. 352(10): p. 987–996. [PubMed: 15758009]
10. Stupp R, et al. , Effects of radiotherapy with concomitant and adjuvant temozolomide versus radiotherapy alone on survival in glioblastoma in a randomised phase III study: 5-year analysis of the EORTC-NCIC trial. *Lancet Oncol*, 2009. 10(5): p. 459–66. [PubMed: 19269895]
11. Stummer W, et al. , Fluorescence-guided surgery with 5-aminolevulinic acid for resection of malignant glioma: a randomised controlled multicentre phase III trial. *Lancet Oncol*, 2006. 7(5): p. 392–401. [PubMed: 16648043]
12. Louveau A, et al. , Structural and functional features of central nervous system lymphatic vessels. *Nature*, 2015. 523(7560): p. 337–41. [PubMed: 26030524]
13. Charles NA, et al. , The brain tumor microenvironment. *Glia*, 2011. 59(8): p. 1169–80. [PubMed: 21446047]
14. Saka M, et al. , Vaccine therapy with dendritic cells transfected with Il13ra2 mRNA for glioma in mice. *J Neurosurg*, 2010. 113(2): p. 270–9. [PubMed: 19895199]
15. Jackson C, et al. , Challenges in immunotherapy presented by the glioblastoma multiforme microenvironment. *Clin Dev Immunol*, 2011. 2011: p. 732413. [PubMed: 22190972]
16. Mahaley MS, et al. , Immunobiology of primary intracranial tumors. *Journal of Neurosurgery*, 1977. 46: p. 467–476. [PubMed: 191575]
17. Brooks W, et al. , Depressed cell-mediated immunity in patients with primary intracranial tumors. *Journal of Experimental Medicine*, 1972. 136: p. 1631–1647. [PubMed: 4345108]
18. Bodmer S, et al. , Immunosuppression and transforming growth factor-beta in glioblastoma. Preferential production of transforming growth factor-beta 2. *The Journal of Immunology*, 1989. 143: p. 3222–3229. [PubMed: 2809198]
19. Wintterle S, et al. , Expression of the B7-Related Molecule B7-H1 by Glioma Cells: A Potential Mechanism of Immune Paralysis. *Cancer Res*, 2003. 63: p. 7462–7467. [PubMed: 14612546]
20. Zhang F, et al. , Genetic programming of macrophages to perform anti-tumor functions using targeted mRNA nanocarriers. *Nat Commun*, 2019. 10(1): p. 3974. [PubMed: 31481662]
21. Fujimoto T, Tomizawa M, and Yokosuka O, SiRNA of frizzled-9 suppresses proliferation and motility of hepatoma cells. *Int J Oncol*, 2009. 35(4): p. 861–6. [PubMed: 19724923]
22. Stafman L, et al. , Targeting PIM kinase as a therapeutic strategy in human hepatoblastoma. *Oncotarget*, 2018. 9(22): p. 22665–22679. [PubMed: 29854306]
23. Soini T, et al. , Downregulation of transcription factor GATA4 sensitizes human hepatoblastoma cells to doxorubicin-induced apoptosis. *Tumor Biology*, 2017. 39(3).
24. Shi Q, et al. , Identification of CDC20 as a Novel Biomarker in Diagnosis and Treatment of Wilms Tumor. *Front Pediatr*, 2021. 9: p. 663054. [PubMed: 34513754]
25. Cao X, et al. , Histone deacetylase 5 promotes Wilms' tumor cell proliferation through the upregulation of c-Met. *Mol Med Rep*, 2016. 13(3): p. 2745–50. [PubMed: 26847592]

26. Maeshima R, et al. , MYCN Silencing by RNAi Induces Neurogenesis and Suppresses Proliferation in Models of Neuroblastoma with Resistance to Retinoic Acid. *Nucleic Acid Ther*, 2020. 30(4): p. 237–248. [PubMed: 32240058]
27. Geng N, et al. , Knockdown of ferroportin accelerates erastin-induced ferroptosis in neuroblastoma cells. *European Review for Medical and Pharmacological Sciences*, 2018. 22: p. 3826–3836. [PubMed: 29949159]
28. Zhang QQ, et al. , LncRNA FOXD2-AS1 knockdown inhibits the resistance of human osteosarcoma cells to cisplatin by inhibiting miR-143 expression. *European Review for Medical and Pharmacological Sciences*, 2021. 25: p. 678–686. [PubMed: 33577022]
29. Shen S, et al. , CircECE1 activates energy metabolism in osteosarcoma by stabilizing c-Myc. *Mol Cancer*, 2020. 19(1): p. 151. [PubMed: 33106166]
30. Masurkar SA, et al. , Downregulation of CRX, a Group 3-specific oncogenic transcription factor, inhibits TGF-beta/activin signaling in medulloblastoma cells. *Biochem Biophys Res Commun*, 2021. 568: p. 76–82. [PubMed: 34192607]
31. Bhatia S, et al. , Knockdown of EphB1 receptor decreases medulloblastoma cell growth and migration and increases cellular radiosensitization. *Oncotarget*, 2015. 6: p. 8929–8946. [PubMed: 25879388]
32. Iegiani G, et al. , CENPE Inhibition Leads to Mitotic Catastrophe and DNA Damage in Medulloblastoma Cells. *Cancers (Basel)*, 2021. 13(5).
33. Yuan X, Sun Z, and Cui C, Knockdown of lncRNA HOTTIP Inhibits Retinoblastoma Progression by Modulating the miR-101–3p/STC1 Axis. *Technol Cancer Res Treat*, 2021. 20: p. 1533033821997831. [PubMed: 33784880]
34. Li Z, et al. , Knockdown of NRMT enhances sensitivity of retinoblastoma cells to cisplatin through upregulation of the CENPA/Myc/Bcl2 axis. *Cell Death Discov*, 2022. 8(1): p. 14. [PubMed: 35013138]
35. Zhang Y, et al. , Knockdown of GPSM1 Inhibits the Proliferation and Promotes the Apoptosis of B-Cell Acute Lymphoblastic Leukemia Cells by Suppressing the ADCY6-RAPGEF3-JNK Signaling Pathway. *Pathol Oncol Res*, 2021. 27: p. 643376. [PubMed: 34257610]
36. Zhang N, et al. , CARMA1 is required for Notch1-induced NF-kappaB activation in SIL-TAL1-negative T cell acute lymphoblastic leukemia. *J Mol Med (Berl)*, 2021. 99(10): p. 1447–1458. [PubMed: 34223928]
37. Sridharan K and Gogtay NJ, Therapeutic nucleic acids: current clinical status. *Br J Clin Pharmacol*, 2016. 82(3): p. 659–72. [PubMed: 27111518]
38. Tang X, et al. , Therapeutic Prospects of mRNA-Based Gene Therapy for Glioblastoma. *Front Oncol*, 2019. 9: p. 1208. [PubMed: 31781503]
39. Meng Z, et al. , A new developing class of gene delivery: messenger RNA-based therapeutics. *Biomater Sci*, 2017. 5(12): p. 2381–2392. [PubMed: 29063914]
40. Tavernier G, et al. , mRNA as gene therapeutic: how to control protein expression. *J Control Release*, 2011. 150(3): p. 238–47. [PubMed: 20970469]
41. Shen T, et al. , Nucleic Acid Immunotherapeutics for Cancer. *ACS Appl Bio Mater*, 2020. 3(5): p. 2838–2849.
42. Bollam SR, Berens ME, and Dhruv HD, When the Ends Are Really the Beginnings: Targeting Telomerase for Treatment of GBM. *Curr Neurol Neurosci Rep*, 2018. 18(4): p. 15. [PubMed: 29525892]
43. Salloum R, et al. , A molecular biology and phase II study of imetelstat (GRN163L) in children with recurrent or refractory central nervous system malignancies: a pediatric brain tumor consortium study. *J Neurooncol*, 2016. 129(3): p. 443–451. [PubMed: 27350411]
44. Cen B, et al. , An Efficient Bivalent Cyclic RGD-PIK3CB siRNA Conjugate for Specific Targeted Therapy against Glioblastoma In Vitro and In Vivo. *Mol Ther Nucleic Acids*, 2018. 13: p. 220–232. [PubMed: 30312846]
45. Besse A, et al. , MiR-338–5p sensitizes glioblastoma cells to radiation through regulation of genes involved in DNA damage response. *Tumour Biol*, 2016. 37(6): p. 7719–27. [PubMed: 26692101]

46. Yoon S, et al. , An RNA Aptamer Targeting the Receptor Tyrosine Kinase PDGFRalpha Induces Anti-tumor Effects through STAT3 and p53 in Glioblastoma. *Mol Ther Nucleic Acids*, 2019. 14: p. 131–141. [PubMed: 30594071]
47. Peng H, et al. , Intracranial delivery of synthetic mRNA to suppress glioblastoma. *Mol Ther Oncolytics*, 2022. 24: p. 160–170. [PubMed: 35024442]
48. Jiao X, et al. , Dual-targeting and microenvironment-responsive micelles as a gene delivery system to improve the sensitivity of glioma to radiotherapy. *Acta Pharm Sin B*, 2019. 9(2): p. 381–396. [PubMed: 30972284]
49. Bertucci A, et al. , Combined Delivery of Temozolomide and Anti-miR221 PNA Using Mesoporous Silica Nanoparticles Induces Apoptosis in Resistant Glioma Cells. *Small*, 2015. 11(42): p. 5687–95. [PubMed: 26395266]
50. Zhang Z, et al. , Nucleic acid-based therapy for brain cancer: Challenges and strategies. *J Control Release*, 2022. 350: p. 80–92. [PubMed: 35970297]
51. Liu J, et al. , Cancer vaccines as promising immuno-therapeutics: platforms and current progress. *J Hematol Oncol*, 2022. 15(1): p. 28. [PubMed: 35303904]
52. Schumacher T, et al. , A vaccine targeting mutant IDH1 induces antitumour immunity. *Nature*, 2014. 512(7514): p. 324–7. [PubMed: 25043048]
53. Guo XR, et al. , PTEN-mRNA engineered mesenchymal stem cell-mediated cytotoxic effects on U251 glioma cells. *Oncol Lett*, 2016. 11(4): p. 2733–2740. [PubMed: 27073544]
54. Schuessler A, et al. , Autologous T-cell therapy for cytomegalovirus as a consolidative treatment for recurrent glioblastoma. *Cancer Res*, 2014. 74(13): p. 3466–76. [PubMed: 24795429]
55. Ashley D, et al. , Bone Marrow-generated Dendritic Cells Pulsed with Tumor Extracts or Tumor RNA Induce Antitumor Immunity against Central Nervous System Tumors. *Journal of Experimental Medicine*, 1997. 186(7): p. 1177–1182. [PubMed: 9314567]
56. Rahman MM, Zhou N, and Huang J, An Overview on the Development of mRNA-Based Vaccines and Their Formulation Strategies for Improved Antigen Expression In Vivo. *Vaccines (Basel)*, 2021. 9(3).
57. Turley EA, et al. , Expression and function of a receptor for hyaluronan-mediated motility on normal and malignant B lymphocytes. *Blood*, 1993. 81(2): p. 446–453. [PubMed: 7678518]
58. Amano T, et al. , Antitumor effects of vaccination with dendritic cells transfected with modified receptor for hyaluronan-mediated motility mRNA in a mouse glioma model. *Journal of Neurosurgery*, 2007. 106: p. 638–645. [PubMed: 17432716]
59. Orange DE, et al. , Effective antigen cross-presentation by prostate cancer patients' dendritic cells: implications for prostate cancer immunotherapy. *Prostate Cancer Prostatic Dis*, 2004. 7(1): p. 63–72. [PubMed: 14999241]
60. Jarboe JS, et al. , Expression of interleukin-13 receptor alpha2 in glioblastoma multiforme: implications for targeted therapies. *Cancer Res*, 2007. 67(17): p. 7983–6. [PubMed: 17804706]
61. Hershey GK, IL-13 receptors and signaling pathways: an evolving web. *J Allergy Clin Immunol*, 2003. 111(4): p. 677–90; quiz 691. [PubMed: 12704343]
62. Parry RG, Gillespie KM, and Mathieson PW, Effects of Type 2 Cytokines on Glomerular Epithelial Cells. *Experimental Nephrology*, 2001. 9: p. 275–283. [PubMed: 11423727]
63. Wilson RJ, et al. , Spatial patterns in species distributions reveal biodiversity change. *Nature*, 2004. 432(7015): p. 393–6. [PubMed: 15549106]
64. Kase M, et al. , Impact of CD133 positive stem cell proportion on survival in patients with glioblastoma multiforme. *Radiol Oncol*, 2013. 47(4): p. 405–10. [PubMed: 24294187]
65. Sayour EJ, et al. , Systemic activation of antigen-presenting cells via RNA-loaded nanoparticles. *Oncoimmunology*, 2017. 6(1): p. e1256527. [PubMed: 28197373]
66. Grippin A, et al. , Versatile mRNA-Nanoparticle Platform for Treatment of Glioblastoma. *International Journal of Radiation Oncology*Biophysics*, 2021. 111(3): p. e251–e252.
67. Kortylewski M, et al. , Inhibiting Stat3 signaling in the hematopoietic system elicits multicomponent antitumor immunity. *Nat Med*, 2005. 11(12): p. 1314–21. [PubMed: 16288283]

68. Hussain SF, et al. , A novel small molecule inhibitor of signal transducers and activators of transcription 3 reverses immune tolerance in malignant glioma patients. *Cancer Res*, 2007. 67(20): p. 9630–6. [PubMed: 17942891]
69. Yaghi NK, et al. , Immune modulatory nanoparticle therapeutics for intracerebral glioma. *Neuro Oncol*, 2017. 19(3): p. 372–382. [PubMed: 27765835]
70. Wei J, et al. , miR-124 inhibits STAT3 signaling to enhance T cell-mediated immune clearance of glioma. *Cancer Res*, 2013. 73(13): p. 3913–26. [PubMed: 23636127]
71. Jarosz-Biej M, et al. , M1-like macrophages change tumor blood vessels and microenvironment in murine melanoma. *PLoS One*, 2018. 13(1): p. e0191012. [PubMed: 29320562]
72. Yuan A, et al. , Opposite Effects of M1 and M2 Macrophage Subtypes on Lung Cancer Progression. *Sci Rep*, 2015. 5: p. 14273. [PubMed: 26399191]
73. Krausgruber T, et al. , IRF5 promotes inflammatory macrophage polarization and TH1-TH17 responses. *Nat Immunol*, 2011. 12(3): p. 231–8. [PubMed: 21240265]
74. Ren J, Chen X, and Chen ZJ, IKKbeta is an IRF5 kinase that instigates inflammation. *Proc Natl Acad Sci U S A*, 2014. 111(49): p. 17438–43. [PubMed: 25326420]
75. Hambardzumyan D, et al. , Modeling Adult Gliomas Using RCAS/t-va Technology. *Transl Oncol*, 2009. 2(2): p. 89–95. [PubMed: 19412424]
76. Gary EN and Weiner DB, DNA vaccines: prime time is now. *Curr Opin Immunol*, 2020. 65: p. 21–27. [PubMed: 32259744]
77. Lopes A, Vandermeulen G, and Preat V, Cancer DNA vaccines: current preclinical and clinical developments and future perspectives. *J Exp Clin Cancer Res*, 2019. 38(1): p. 146. [PubMed: 30953535]
78. Lopes A, et al. , New generation of DNA-based immunotherapy induces a potent immune response and increases the survival in different tumor models. *J Immunother Cancer*, 2021. 9(4).
79. O, I., et al. , A DNA vaccine expressing tyrosinase-related protein-2 induces T-cell-mediated protection against mouse glioblastoma. *Cancer Gene Ther*, 2003. 10(9): p. 678–88. [PubMed: 12944987]
80. Chi D, et al. , Molecular Detection of Tumor-Associated Antigens Shared by Human Cutaneous Melanomas and Gliomas. *American Journal of Pathology*, 1997. 150(6): p. 2143–2152. [PubMed: 9176405]
81. Nigro JM, et al. , Scrambled Exons. *Cell*, 1991. 64: p. 607–613. [PubMed: 1991322]
82. Li Y, et al. , Circular RNA is enriched and stable in exosomes: a promising biomarker for cancer diagnosis. *Cell Res*, 2015. 25(8): p. 981–4. [PubMed: 26138677]
83. Li J, et al. , Long Noncoding RNAs Regulate Cell Growth, Proliferation, and Apoptosis. *DNA Cell Biol*, 2016. 35(9): p. 459–70. [PubMed: 27213978]
84. Liu X, et al. , Synthetic Circular RNA Functions as a miR-21 Sponge to Suppress Gastric Carcinoma Cell Proliferation. *Molecular Therapy - Nucleic Acids*, 2018. 13: p. 312–321. [PubMed: 30326427]
85. Wang Z, et al. , Synthetic circular multi-miR sponge simultaneously inhibits miR-21 and miR-93 in esophageal carcinoma. *Lab Invest*, 2019. 99(10): p. 1442–1453. [PubMed: 31217510]
86. Goodall GJ and Wickramasinghe VO, RNA in cancer. *Nat Rev Cancer*, 2021. 21(1): p. 22–36. [PubMed: 33082563]
87. Wang R, et al. , EIF4A3-induced circular RNA MMP9 (circMMP9) acts as a sponge of miR-124 and promotes glioblastoma multiforme cell tumorigenesis. *Mol Cancer*, 2018. 17(1): p. 166. [PubMed: 30470262]
88. Zhou J, et al. , Circular RNA hsa_circ_0008344 regulates glioblastoma cell proliferation, migration, invasion, and apoptosis. *J Clin Lab Anal*, 2018. 32(7): p. e22454. [PubMed: 29687495]
89. Li X and Diao H, Circular RNA circ_0001946 acts as a competing endogenous RNA to inhibit glioblastoma progression by modulating miR-671–5p and CDR1. *J Cell Physiol*, 2019. 234(8): p. 13807–13819. [PubMed: 30663767]
90. Wei Y, et al. , EIF4A3-induced circular RNA ASAP1 promotes tumorigenesis and temozolomide resistance of glioblastoma via NRAS/MEK1/ERK1–2 signaling. *Neuro Oncol*, 2021. 23(4): p. 611–624. [PubMed: 32926734]

91. Chen B, et al. , Circular RNA circLGMN facilitates glioblastoma progression by targeting miR-127-3p/LGMN axis. *Cancer Lett*, 2021. 522: p. 225–237. [PubMed: 34582975]
92. Yang P, et al. , Silencing of cZNF292 circular RNA suppresses human glioma tube formation via the Wnt/ β -catenin signaling pathway. *Oncotarget*, 2016. 7(39): p. 63449–63455. [PubMed: 27613831]
93. Zhu J, et al. , Differential Expression of Circular RNAs in Glioblastoma Multiforme and Its Correlation with Prognosis. *Transl Oncol*, 2017. 10(2): p. 271–279. [PubMed: 28236760]
94. Wu X, et al. , A novel protein encoded by circular SMO RNA is essential for Hedgehog signaling activation and glioblastoma tumorigenicity. *Genome Biol*, 2021. 22(1): p. 33. [PubMed: 33446260]
95. Gao X, et al. , Circular RNA-encoded oncogenic E-cadherin variant promotes glioblastoma tumorigenicity through activation of EGFR-STAT3 signalling. *Nat Cell Biol*, 2021. 23(3): p. 278–291. [PubMed: 33664496]
96. Zhong J, et al. , Circular EZH2-encoded EZH2-92aa mediates immune evasion in glioblastoma via inhibition of surface NKG2D ligands. *Nat Commun*, 2022. 13(1): p. 4795. [PubMed: 35970825]
97. Sun W, et al. , Circular RNA: A novel type of biomarker for glioma (Review). *Mol Med Rep*, 2021. 24(2).
98. Kristensen LS, et al. , The emerging roles of circRNAs in cancer and oncology. *Nat Rev Clin Oncol*, 2022. 19(3): p. 188–206. [PubMed: 34912049]
99. Muller S, et al. , Synthetic circular miR-21 RNA decoys enhance tumor suppressor expression and impair tumor growth in mice. *NAR Cancer*, 2020. 2(3): p. zcaa014. [PubMed: 34316687]
100. Chen C and Sarnow P, Initiation of Protein Synthesis by the Eukaryotic Translational Apparatus on Circular RNAs. *Science*, 1995. 268(5209): p. 415–417. [PubMed: 7536344]
101. Qu L, et al. , Circular RNA vaccines against SARS-CoV-2 and emerging variants. *Cell*, 2022. 185(10): p. 1728–1744 e16. [PubMed: 35460644]
102. Li H, et al. , Circular RNA cancer vaccines drive immunity in hard-to-treat malignancies. *Theranostics*, 2022. 12(14): p. 6422–6436. [PubMed: 36168634]
103. Wesselhoeft RA, Kowalski PS, and Anderson DG, Engineering circular RNA for potent and stable translation in eukaryotic cells. *Nat Commun*, 2018. 9(1): p. 2629. [PubMed: 29980667]
104. Kulkarni JA, et al. , The current landscape of nucleic acid therapeutics. *Nat Nanotechnol*, 2021. 16(6): p. 630–643. [PubMed: 34059811]
105. Chen J, et al. , Lipid nanoparticle-mediated lymph node-targeting delivery of mRNA cancer vaccine elicits robust CD8(+) T cell response. *Proc Natl Acad Sci U S A*, 2022. 119(34): p. e2207841119. [PubMed: 35969778]
106. Nabhan JF, et al. , Intrathecal delivery of frataxin mRNA encapsulated in lipid nanoparticles to dorsal root ganglia as a potential therapeutic for Friedreich's ataxia. *Sci Rep*, 2016. 6: p. 20019. [PubMed: 26883577]
107. Ma F, et al. , Neurotransmitter-derived lipidoids (NT-lipidoids) for enhanced brain delivery through intravenous injection. *Science Advances*, 2020. 6.
108. Anraku Y, et al. , Glycaemic control boosts glucosylated nanocarrier crossing the BBB into the brain. *Nat Commun*, 2017. 8(1): p. 1001. [PubMed: 29042554]
109. Yang L, et al. , Extracellular Vesicle-Mediated Delivery of Circular RNA SCMH1 Promotes Functional Recovery in Rodent and Nonhuman Primate Ischemic Stroke Models. *Circulation*, 2020. 142(6): p. 556–574. [PubMed: 32441115]
110. Carbonaro TM and Gatch MB, Neuropharmacology of N,N-dimethyltryptamine. *Brain Research Bulletin*, 2016. 126: p. 74–88. [PubMed: 27126737]
111. Hargadon KM, Johnson CE, and Williams CJ, Immune checkpoint blockade therapy for cancer: An overview of FDA-approved immune checkpoint inhibitors. *Int Immunopharmacol*, 2018. 62: p. 29–39. [PubMed: 29990692]

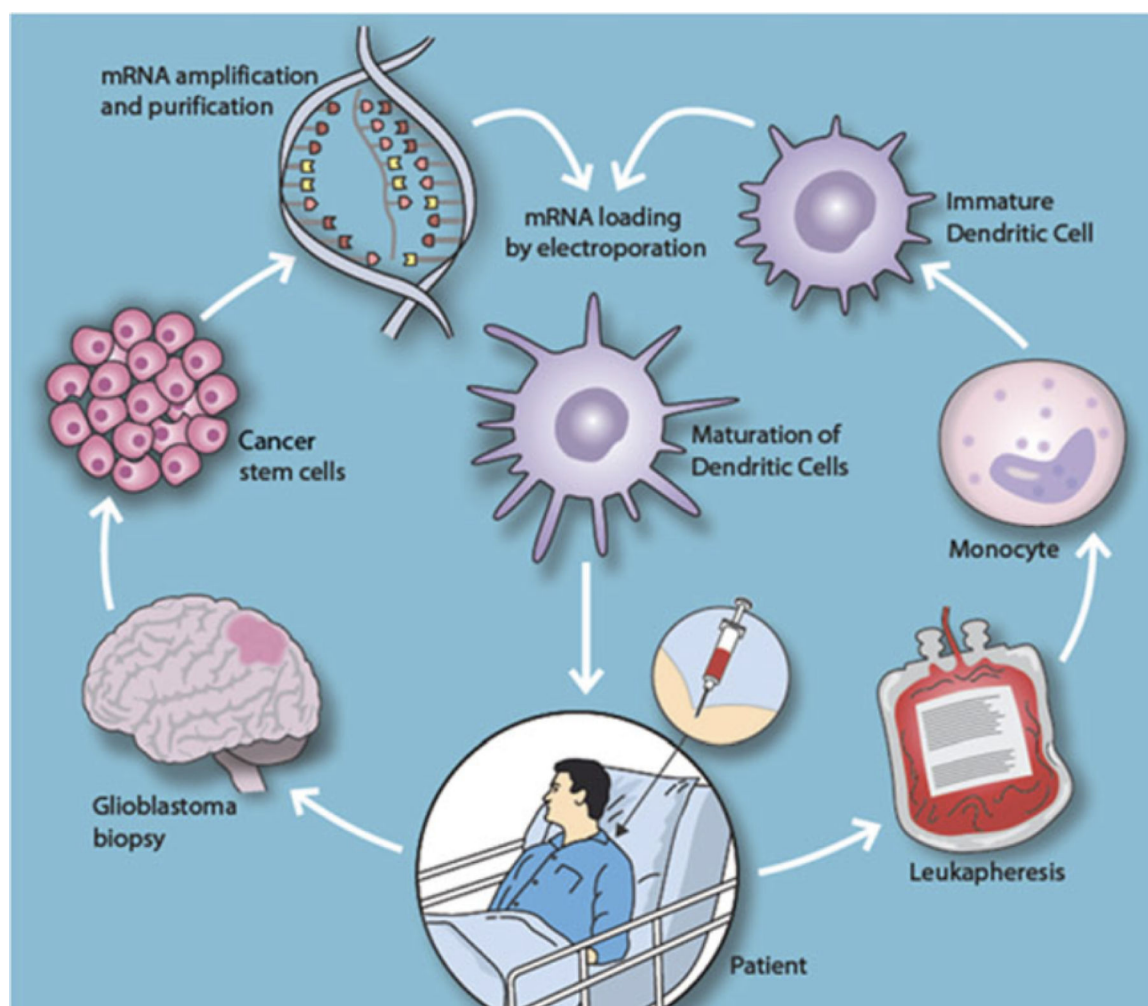


Figure 1. Schematic illustration of the development of autologous mRNA DC vaccines [2]. *Left:* tumor cells were collected during surgery and cultured to expand brain cancer stem cells. mRNA was then purified from the stem cell culture and amplified. *Right:* monocytes were collected from patients via leukapheresis before the start of radio- and chemo-therapy, which were then ex-vivo cultured and differentiated into immature DCs. *Center:* DCs were pulsed with autologous brain cancer stem cell mRNA, creating the completed DC vaccines and administered to patients upon the completion of radio- and chemo-therapy.

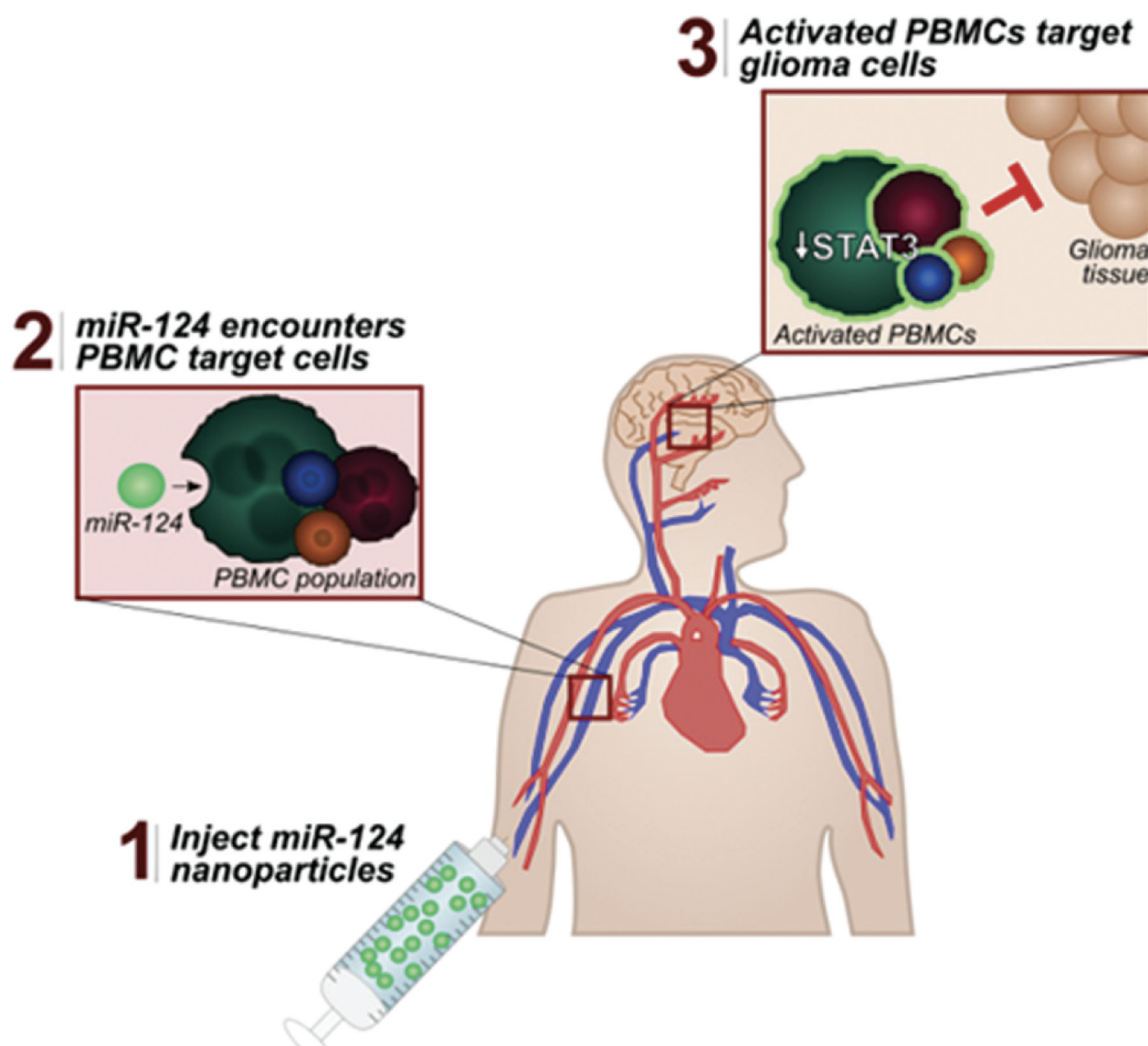


Figure 2.

Proposed mechanism of action of the miR-124 NPs (LUNAR-301) [69]. LUNAR-301 is first administered intravenously. In the peripheral blood, LUNAR-301 is taken up by PBMCs and releases miR-124 contents in response to the acidic endosomes, followed by endosome escape. Once in the cytoplasm of the PBMCs, miR-124 inhibits the translation of STAT3 which allows the PBMCs to act as APCs and activate T cells. Activated T cells can then target the glioma cells.

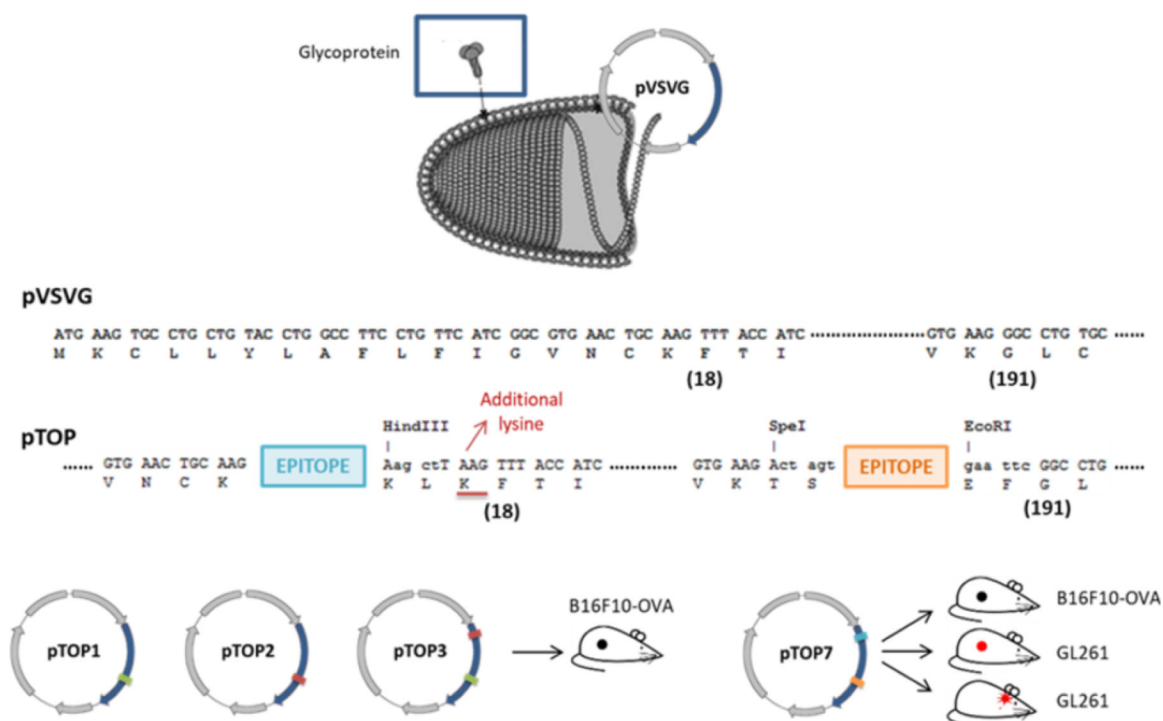


Figure 3.

Schematic illustration of a plasmid that encodes T cell epitopes (pTOP) [78]. pVSVG refers to the native plasmid sequence before alteration. pTOP illustrates where the foreign epitopes were inserted into pVSVG via digestion of restriction sites by BamHI and HindIII at position 18 or by SpeI and EcoRI at position 191. Several versions of pTOP can be created by mixing and matching different epitopes at either or both insertion sites.

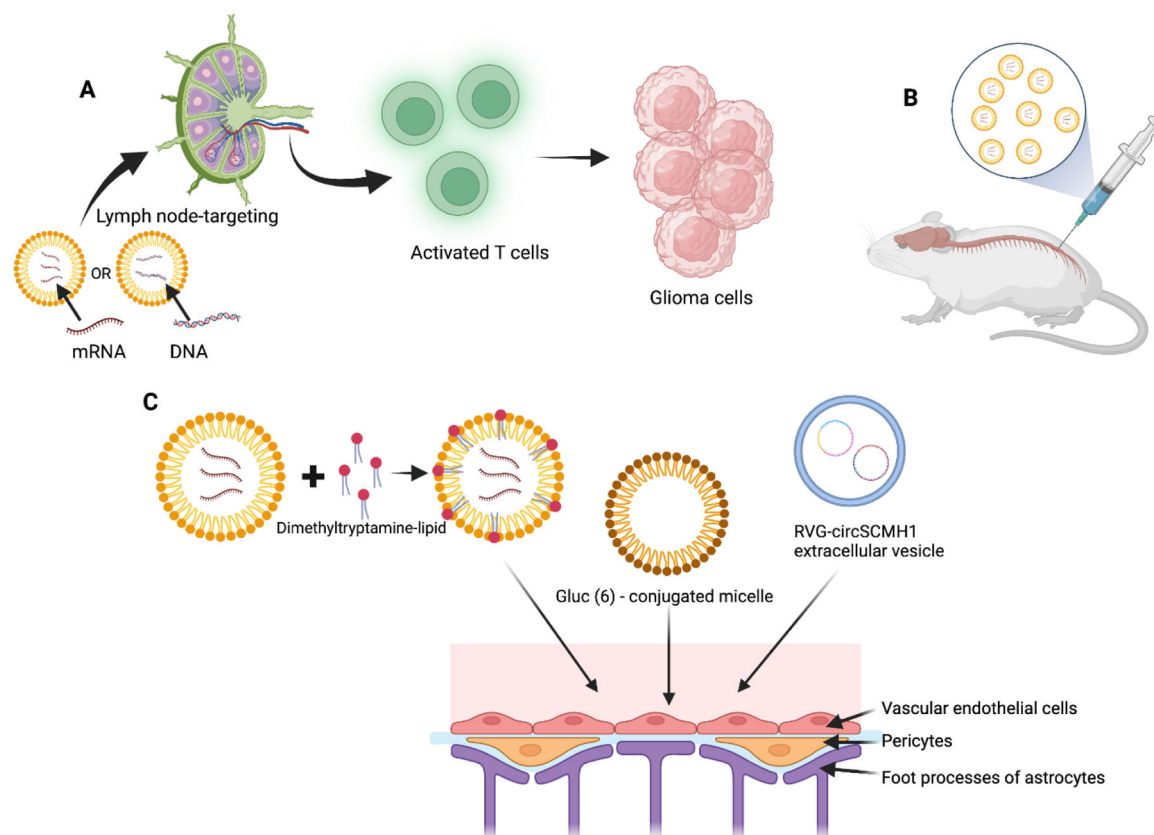


Figure 5.

Summary of representative delivery methods for nucleic acid therapeutics and vaccines for potential application in GBM therapy. **A.** mRNA or DNA can be packaged into LNPs which then target lymph nodes to interact with naïve T cells, which become activated and can target glioma tissue. **B.** Intrathecal delivery of frataxin mRNA into the spinal column to target DRG [106]. **C.** Dimethyltryptamine-conjugated LNPs [107], Gluc(6)-conjugated micelles [108], and RVG-circSCMH1 extracellular vesicles [109] have all been found to target and pass through the BBB to deliver their payloads at the site of disease within the CNS. Figure was created with BioRender.

Name	Epitope inserted in position 18	Epitope inserted in position 191
pVSVG	-	-
pTOP1	-	OVA257–264 SIINFEKL
pTOP2	-	OVA323–339 ISQAVHAAHAEINEAGR
pTOP3	OVA323–339 ISQAVHAAHAEINEAGR	OVA257–264 SIINFEKL
pTOP7	gp10044–59 WNRQLYPEWTEAQLD	TRP2180–188 SVYDFFVWL

Table 1.

Summary of nucleic acid immunotherapeutics for GBM.

Type	Mechanism	Reference
mRNA-pulsed dendritic cell (DC) vaccines	Whole-cell or tumor-associated antigen presentation by dendritic cells to CD4 ⁺ and CD8 ⁺ cells in the lymph nodes.	[2, 4, 14, 55, 58]
RNA-loaded nanoparticles targeting tumor cells	RNA-loaded nanoparticle complexes activate T cells in the periphery.	[65, 66]
RNA-loaded nanoparticles targeting systemic immune system	RNA-lipid nanoparticle (LNP) complexes target peripheral blood mononuclear cells.	[69]
RNA-loaded nanoparticles targeting macrophages	Nanoparticles carrying mRNA that code M1-polarizing transcription factors target tumor-associated macrophages.	[20]
DNA vaccines	DNA encoding tumor TRP-2 _{180–188} and gp100 _{44–59} epitopes target and activate T cells.	[78]

Table 2.

Summary of TRP-2_{180–188} and gp100_{44–59} epitopes inserted into positions 18 and 191 of pVSVG to create pTOP1, 2, 3, and 7 [78].

Name	Epitope inserted in position 18		Epitope inserted in position 191	
pVSVG	-		-	
pTOP1	-		OVA _{257–264}	SIINFEKL
pTOP2	-		OVA _{323–339}	ISQAVHAAHAEINEAGR
pTOP3	OVA _{323–339}	ISQAVHAAHAEINEAGR	OVA _{257–264}	SIINFEKL
pTOP7	gp100 _{44–59}	WNRQLYYPEWTEAQLD	TRP _{180–188}	SVYDFFVWL